Farmora - Comprehensive Academic Report (Milestones 1 to 6)

Department of Software Engineering, Faculty of Computing
Sri Lanka Institute of Information Technology (SLIIT)
Mobile Application Project - Complete Academic Dossier

SLIIT Department of Software Engineering - Faculty of Computing

Mobile Application Project: Farmora

Milestone 1: UX Team Formation, User Research & Key Flows (LO1, LO2)

1. Project Title & Overview

Project Name: Farmora ? AI-Powered Smart Agricultural Ecosystem & Traceable Supply Chain
Domain: AgriTech / Supply Chain & Digital Marketplaces
Target Region: Sri Lanka (Western, Central, North-Western & Northern Provinces)
App Type: Cross-Platform Mobile & Progressive Web Application (Flutter + Firebase Serverless)

2. UX Team Formation & Roles (Lab 01 & Lab 02)

To satisfy the semester lab requirements, team members took ownership of distinct user experiences:

Member Name	Student ID	Designated UX Role	Focus Modules & Responsibiliti
Ahsan Mohammed	ITXXXXXXX (Lead)	Lead Architect & Admin/Buyer U	Overall system architecture, B
Kajana S.	ITXXXXXXX	Farmer Operations & Agronomy R	Farmer registration, Harvest l
Sureka (Suka)	ITXXXXXXX	Logistics & Transporter UX	Transporter dispatching, Fleet
Team Member 4	ITXXXXXXX	UX Research & QA Lead	Usability benchmarking, Access

3. User Groups & Persona Identification (Lab 03)

Persona 1: Sunimal Perera (Smallholder Farmer - Dambulla)

- Age:** 48
- Context:** Operates a 2-acre vegetable farm; experiences 25-35% post-harvest spoilage and price exploitation by middlemen at economic centres.
- Digital Literacy:** Low-to-Medium (WhatsApp, YouTube, voice notes; struggles with complex text menus).
- Core Goals:** Fair market pricing without broker cartels; rapid harvest listing; agronomic disease detection without costly consultants.

- App Needs:** Simple 1-tap harvest logging, voice search, high-contrast visual cues, Sinhala/Tamil support, QR code traceability stickers.

Persona 2: Sarah Mendis (Supermarket/Hotel Food Procurement Manager - Colombo)

- Age:** 34
- Context:** Procures 1,500kg of fresh vegetables daily for retail distribution; requires grade-A consistency, timely delivery, and farm-to-shelf traceability.
- Digital Literacy:** High (Enterprise ERPs, mobile banking, inventory apps).
- Core Goals:** Verified quality grading, transparent cold-chain verification, escrow-protected payments, automated VAT invoicing.
- App Needs:** Bulk ordering, real-time dispatch tracking, digital certificates, escrow release on quality verification.

Persona 3: Ruwan Bandara (Independent Fleet Transporter - Kurunegala)

- Age:** 41
- Context:** Owns 2 refrigerated trucks (Reefers) and 1 light lorry; frequently returns empty on return trips (deadhead miles).
- Digital Literacy:** Medium (Google Maps, phone calls, SMS).
- Core Goals:** Optimize load capacity, minimize empty backhauls, monitor reefer temperature to prevent cargo rejection.
- App Needs:** Real-time cargo matching, clear turn-by-turn routing, instant digital Proof-of-Delivery (photo + signature).

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4. User Research Methodology & Findings (Lab 04)

Research Methodology:

- Semi-structured Interviews:** 12 participants (5 Farmers in Dambulla/Nuwara Eliya, 4 Buyers/Wholesalers in Colombo/Pettah, 3 Transporters).
- Contextual Inquiry:** Observed harvest packing and middleman negotiation at the Dambulla Dedicated Economic Centre.
- Survey:** Distributed to 35 agricultural cooperative members via mobile forms.

Key Pain Points Discovered:

1. **Opaque Pricing:** 83% of farmers reported receiving up to 40% less than Colombo retail price due to multi-tiered intermediaries.
2. **Cold-Chain Spoilage:** Over 28% of perishable produce degrades due to lack of temperature logging during transit.
3. **Payment Defaults:** Buyers and farmers experience trust deficits; farmers fear delayed payments, buyers fear substandard rot.
4. **Language Barrier:** Most agricultural apps are English-only, completely alienating local farmers.

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5. Validated Key User Flows (Lab 05)

Flow 1: Farmer Harvest Onboarding & Video Quality QR Generation

‘[Farmer Log In]‘ ? ‘[Capture Harvest Photos/Video]‘ ? ‘[Auto-detect Grade &

Price Guide]‘ ? ‘[Publish Batch]‘ ? ‘[Generate Batch QR Sticker]‘ ? ‘[Awaiting Buyer Purchase]‘

Flow 2: Buyer Direct Purchase & Escrow Settlement

‘[Browse Verified Marketplace]‘ ? ‘[Inspect Video/Batch Audit]‘ ? ‘[Add to Cart & Select Delivery Option]‘ ? ‘[Deposit to PayHere/Escrow]‘ ? ‘[Real-Time Tracking]‘ ? ‘[Inspect at Delivery & Scan QR]‘ ? ‘[Release Escrow Funds to Farmer]‘

Flow 3: Transporter Fleet Allocation & Delivery POD

‘[Dispatch Notification Received]‘ ? ‘[Select Vehicle & Driver]‘ ? ‘[Accept Route with Cold-Storage Parameters]‘ ? ‘[GPS Live Tracking]‘ ? ‘[Upload Delivery Photo & Recipient Signature]‘ ? ‘[Instant Payout Credited]‘

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6. Sketching & Low-Fidelity Wireframes (Lab 06)

Low-fidelity sketches and information hierarchy were designed covering:
1. **Adaptive Role-Based Dashboard:** Tailored navigation for Farmer, Buyer, Transporter, and Agri-Admin.
2. **Accessible Form Layouts:** Large touch targets (min 48dp), clear step progression, and voice input integration.
3. **Marketplace Matrix:** Card-based browse layout displaying grade, harvest timestamp, location, and verified badges.
4. **Fleet Telematics View:** Combined map and cold-storage temperature telemetry widget.

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Milestone 2: Interactive High-Fidelity Prototyping & Design System (LO3)

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1. Prototype Scope & Objectives

The primary objective of Milestone 2 (Lab 07) is to transform low-fidelity wireframes into an interactive, high-fidelity digital prototype adhering strictly to Material Design 3 (M3) and agricultural usability principles.

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2. Design System & Style Guide

2.1 Color Palette

- Primary Brand (Agri Green):** ‘#2E7D32‘ (Represents vitality, nature, harvest prosperity)
- Primary Container / Accent:** ‘#E8F5E9‘ (Soft contrast for card backgrounds)
- Secondary (Harvest Gold/Orange):** ‘#F57C00‘ (Highlights alerts, urgency, pending escrow)

- Neutral Dark (Text/Headers):** '#1E293B' (High legibility on sunlight screens)
- Neutral Light (Surface):** '#F8FAFC' (Clean, non-distracting backgrounds)
- Danger / Temperature Alert:** '#D32F2F' (Cold storage breaches, expired verification)

2.2 Typography

- Primary English/Latin:** Google Inter & Poppins (Weights: 400 Regular, 500 Medium, 600 SemiBold, 700 Bold)
- Sinhala Localization:** Noto Sans Sinhala (Full Unicode rendering support)
- Tamil Localization:** Noto Sans Tamil (Full Unicode rendering support)

2.3 Micro-interactions & Haptic Feedback

- Pull-to-refresh:** Custom leaf-sprout animation indicating live market price updates.
- Haptic Click:** Confirms barcode scanning and camera shutter actions in outdoor farm conditions.
- Dynamic Chips:** Status indicators for 'Order Status' ('Pending', 'In-Transit', 'Delivered', 'Disputed').

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3. High-Fidelity Screen Architecture & Key Interactions

3.1 Onboarding & Multi-Role Authentication

- Seamless phone OTP login alongside biometric (Fingerprint/Face ID).
- Dedicated role switcher: **Farmer**, **Wholesale Buyer**, **Transporter / Fleet Owner**, and **Agri Expert**.
- Role-specific color accents to avoid cognitive confusion when switching roles.

3.2 Farmer Hub & Smart Harvest Publishing

- Clean, stepped wizard for listing crops.
- Real-time video recording & Cloudinary video hosting integration.
- Barcode/QR generator rendering unique traceability hash 'FARM-BATCH-<ID>'.
- Live farm weather widget with micro-climate rain alerts.

3.3 Buyer Direct Marketplace

- Category carousel: *Leafy Greens*, *Root Crops*, *Fruits*, *Spices*, *Export Quality*.
- Product detail sheet featuring interactive video inspection: buyers preview harvest freshness recorded directly in the field before purchase.
- Escrow purchase breakdown: Commodity Price + Logistics Cost + Quality Guarantee Insurance.

3.4 Logistics Fleet Hub & Telematics

- Interactive Google Maps view rendering vehicle markers, depot pins, and transit polylines.
- Real-time Cold Storage Temperature Gauge (Target 2℃ - 7℃) with automated breach alert.
- Fleet manager bottom sheets to dynamically configure Branch Depots, Vehicles, and Drivers.

4. Accessibility & Inclusive Design (WCAG 2.1 AA Compliance)

- Contrast Ratios:** Text-to-background contrast exceeds 4.8:1 across all primary screens for outdoor visibility in Sri Lankan sunlight.
- Minimum Tap Targets:** All buttons, FABs, and dropdown triggers comply with the 48x48dp minimum physical touch area.
- Screen Reader Support:** Semantic labels applied to all icon buttons and graphic status indicators.
- Multi-lingual Switcher:** Instant runtime switching between Sinhala, Tamil, and English with zero app reloads.

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Milestone 3: User Testing, Security Checks & Iteration (LO5)

1. Test Plan & Objectives (Lab 08)

The usability test evaluated whether farmers, buyers, and transporters could accomplish core workflows without assistance, measured error frequency, validated task completion times, and ensured zero security vulnerabilities exist in data exchange.

Test Parameters:

- Participant Count:** 8 participants (3 Farmers, 3 Wholesale Buyers, 2 Transporters).
- Format:** Moderated Usability Testing via Think-Aloud Protocol followed by the standardized System Usability Scale (SUS) questionnaire.
- Environment:** Physical field testing in Dambulla wholesale hub & simulated office environments in Colombo.

2. Test Tasks & Completion Rates

Task ID	Task Description	Target User	Completion Rate	Avg Time (Sec)	Error Rate
T01	Register account & verify role	All	100%	34s	0%
T02	Add a new crop product with vi	Farmer	87.5%	78s	12.5%
T03	Browse marketplace, locate	Buyer	100%	52s	0%
T04	Set up Branch Depot and regist	Transporter	87.5%	68s	12.5%
T05	Confirm order pickup and upda	Transporter	100%	41s	0%

3. System Usability Scale (SUS) Results

The standardized 10-item SUS questionnaire was administered to all 8 participants.

SUS Score Calculation:

- Average SUS Score:** **84.5 / 100**
- Grade Equivalent:** **Grade A (Excellent Usability)**
- Adjective Rating:** Highly acceptable; exceeds the industry benchmark average of 68.0.

Key Qualitative Feedback:

- "I like that I can take a 10-second video of my tomatoes right from the tree. Buyers immediately trust the quality."* ? Farmer, Dambulla.
- "The temperature tracking during delivery gives us complete confidence that leafy greens won't arrive spoiled."* ? Supermarket Buyer, Colombo.

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4. Secure Design & Data Privacy Checks

4.1 Authentication Flow Security:

- Enforced Firebase Phone Auth with reCAPTCHA verification to eliminate automated bot abuse.
- Biometric lock encryption enabled for financial wallet & escrow release screens.

4.2 Cloud Firestore Security Rules Verification:

- Role-Based Access Control (RBAC): Farmers can only mutate their own crop batches (‘request.auth.uid == resource.data.farmerId‘).
- Transporters can only modify assigned shipment states; financial escrow triggers require multi-signature approval.
- Verified that arbitrary document injections are rejected at the Firestore engine level.

4.3 Data Privacy & Media Storage:

- Sensitive proof-of-delivery photos and harvest videos are stored in authenticated storage buckets with expirable tokenized URLs.
- Sensitive payment tokens handled strictly via PCI-DSS compliant payment gateways (PayHere).

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5. Iterations & Improvements (Before vs. After)

Iteration 1: Empty Branch Depot Validation in Fleet Hub

- Observed Pain Point:** When transporters attempted to add a vehicle before creating a depot branch, the app threw a silent ‘StateError (No element)‘.
- Iteration Applied:** Added an early check ‘if (_branches.isEmpty)‘ that provides an immediate, informative snackbar message: *"Please add a Branch Depot first"* and guides them directly to branch registration.

Iteration 2: Harvest Video Auto-Deletion Upon Status Completion

- Observed Pain Point:** Farmers and system storage accumulated large video files for crops that were already sold and consumed.
- Iteration Applied:** Implemented an automated cleanup cycle where harvest videos uploaded for traceability are archived or removed from costly hot storage once the order reaches ‘Delivered & Escrow Released‘, preserving only the immutable batch QR audit hash.

Iteration 3: Low-Connectivity Field Caching

- Observed Pain Point:** Intermittent 4G cellular drops in remote rural farmlands caused form submission loss.
- Iteration Applied:** Enabled Firestore offline persistence (‘enablePersistence()‘) and client-side photo queueing, allowing farmers to capture harvests offline with automatic background sync upon re-establishing connection.

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Milestone 4: Tech Stack Justification & Mobile App Implementation (LO1, LO4)

Part 1: Tech Stack Justification

1. Architectural Options Comparison

Evaluation Criteria	Option A: Flutter (Selected)	Option B: React Native	Option C: Native (Kotlin + Swi
Rendering Performance	**High** (Skia/Impeller direct	**Medium** (JavaScript bridge	**Highest** (Native platform t
Multi-Platform Support	**Unified Codebase** (Android,	**High** (Mobile only; Web req	**None** (Requires 2 entirely
Team Skillset & Velocity	**Rapid** (Hot reload, typed D	**Moderate** (Requires reconci	**Slow** (Split team between S
Accessibility (a11y)	**Built-in** (Rich ‘Semantics‘	**Good** (Requires third-party	**High** (Native a11y APIs)
Hardware & Camera APIs	**Robust** (Direct Camera, QR	**Moderate** (Bridge lag can c	**High** (Full native access)
Development Cost	**Lowest** (1 codebase, 1 buil	**Moderate** (Bridge debugging	**Highest** (Requires distinct

2. Tech Stack Justification Rationale:

Flutter was selected because AgriTech users in Sri Lanka use diverse Android devices (from budget entry-level smartphones to modern flagships), alongside wholesale enterprise buyers accessing via desktop/tablet web browsers. Flutter provides:
1. **Identical Pixel-Perfect UI:** Zero layout discrepancies across different screen densities and aspect ratios.
2. **Offline-Ready Client Architecture:** Integrates seamlessly with Cloud Firestore offline caches and local SQLite/secure storage.
3. **High Productivity:** Hot Reload enabled the team to iterate rapidly during user testing in Milestone 3.

Part 2: Implementation & Folder Architecture

1. Folder Structure (Clean Architecture Pattern)

‘‘‘
lib/
??? core/
? ??? constants/ # AppColors, ApiEndpoints, Strings
? ??? network/ # Http & Cloudinary API clients
? ??? services/ # FirebaseAuthService, GeolocationService
? ??? theme/ # Material 3 light & dark themes
? ??? utils/ # Formatters, barcode helpers, image compression
??? features/
? ??? admin/ # Escrow dispute management, user approvals
? ??? auth/ # OTP verification, role switcher, onboarding
? ??? buyer/ # Catalog, video inspection, checkout, escrow
? ??? farmer/ # Harvest listing, disease detection, QR generator
? ??? transporter/ # Fleet Hub, depot manager, GPS tracking, POD
??? main.dart # Multi-provider initialization, route registry
‘‘‘

2. Version Control & Git Strategy

- Repository:** Hosted on GitHub (‘AHSANMOHAMMED/Farmora-App‘).
- Branching Model:** Feature-branch workflow with automated PR verification and strict branch protection on ‘main‘ and ‘dev/swami‘.
- Release Automation:** Continuous integration building automated Android APKs (‘app-release.apk‘) and continuous deployment to Firebase Web Hosting (‘farmingapp-24b34.web.app‘).

3. Requirements Traceability Matrix (RTM)

Req ID	Requirement / User Story	Implementation Screen / Widget	Data Model / Backend Endpoint	Status
REQ-01	Multi-role user registration	‘welcome_screen.dart‘, ‘role_	‘users/{uid}‘ (role, isVerifie	**Verified**
REQ-02	Farmers can record video ‘&‘ up	‘product_screen.dart‘	‘crops/{cropId}‘ + Cloudinary	**Verified**
REQ-03	Generate unique batch QR for	‘batch_qr_dialog.dart‘, ‘qr_	‘qrCodeHash‘, ‘storagePath‘	**Verified**
REQ-04	Buyers inspect video and ‘pro	‘product_detail_screen.dart‘,	‘orders/{orderId}‘ (escrowSta	**Verified**
REQ-05	Transporters manage branded	‘logistics_fleet_hub_screen.da	‘transporters/{tid}/vehicles‘,	**Verified**
REQ-06	Real-time Cold Storage telem	‘logistics_fleet_map_view.dart	‘cold_storage_logs‘, ‘current	**Verified**
REQ-07	Proof-of-delivery (photo + ‘sig	‘order_dispatch_screen.dart‘,	‘orders/{orderId}/podPhotoPath	**Verified**

4. Build Deliverables

- Mobile Build:** Android Release APK located at ‘build/app/outputs/flutter-apk/app-release.apk‘ (Size: 99.3MB, compileSdk: 36, Java 21 LTS).
- Web App URL:** https://farmingapp-24b34.web.app
- Git Repository:** https://github.com/AHSANMOHAMMED/Farmora-App.git

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Milestone 5: Promotion, Business Pitching & Marketing Strategy (LO6)

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1. Value Proposition Canvas & Positioning Statement

1.1 Value Proposition Canvas

“
??
? THE CUSTOMER ? THE VALUE MAP ?
??
? Customer Jobs: ? Products & Services: ?
? ? Sell harvest at profitable rates ? ? Farmora Mobile App (Flutter) ?
? ? Source fresh, high-grade produce ? ? Farm-to-Fork QR Traceability ?
? ? Deliver cargo without empty miles ? ? Smart Escrow Payment Engine ?
? ? ?
? Pains: ? Pain Relievers: ?
? ? Middleman exploitation (up to 40%) ? ? Direct marketplace eliminates cuts ?
? ? Perishable cold-chain spoilage ? ? Real-time reefer temperature alert ?
? ? Payment default and bad debts ? ? Automated escrow guarantees funds ?
? ? ?
? Gains: ? Gain Creators: ?
? ? Higher profit margins for farmers ? ? Guaranteed 48-hr direct payouts ?
? ? Certified quality for retail buyers? ? Video proof of harvest freshness ?
? ? Optimized transporter vehicle runs ? ? Multi-drop route matching engine ?
??
“

1.2 Positioning Statement

> *"For commercial agricultural producers, wholesale buyers, and logistics operators in Sri Lanka who are burdened by predatory middlemen and cold-chain losses, **Farmora** is an end-to-end digital ecosystem that provides transparent price discovery, video-verified crop quality, and escrow-protected logistics. Unlike traditional physical economic centres and fragmented broker cartels, Farmora guarantees fair income, zero default risk, and verified food safety directly from farm to retail."*

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2. Go-to-Market (GTM) Strategy

Phase 1: Pilot Launch & Grassroots Activation (Months 1?3)

- Target Hub:** Dambulla and Nuwara Eliya agricultural belts.
- On-the-ground Farmer Ambassadors:** Partner with local Agrarian Service Centres and tea/vegetable farmer co-operatives to onboard the first 250 farmers.

- Incentive:** 0% commission on the first 5 transactions + free batch QR sticker printer access at local depots.

Phase 2: Wholesale Buyer & Supermarket Integration (Months 4?6)

- Target Buyers:** Supermarket chains (Keells, Cargills, Arpico), major hotels, and Colombo wholesale catering syndicates.
- Value Delivery:** Dedicated procurement dashboard with bulk invoicing and automated cold-chain audit certificates.

Phase 3: Regional Expansion & Logistics Network (Months 7?12)

- Expansion into Jaffna, Anuradhapura, and Matale districts.
- Full commercial rollout of the Transporter Fleet Hub for cross-island cold chain transport.

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3. Business & Monetization Model

Farmora operates on a sustainable multi-stream revenue model:

Revenue Stream	Model Type	Pricing Structure	Target Segment
**Marketplace Transaction Fee*	Commission	2.5% per completed wholesale o	Wholesale Buyers
Escrow & Quality Guarantee	Escrow Assurance	1.0% escrow protection fee ens	All Marketplace Users
Logistics Freight Matching	Platform Fee	3.5% per dispatch trip matched	Transporters
Enterprise SaaS Dashboard	Subscription	$25/month for multi-branch war	Supermarkets & Exporters
Agri-Fintech Micro-loans	Partner Revenue Share	1.2% origination fee on micro-	Farmers

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4. Emerging UX Trends Tie-In

- AI-Powered Harvest Visual Inspection:** Computer vision analyzes uploaded harvest photos to recommend initial quality grading (Grade A/B/C) and alert farmers if leaf discoloration suggests blight.
- Voice UX & Conversational Local Dialects:** Integrated natural-language voice input enabling farmers in rural fields to search market prices and list products hands-free in colloquial Sinhala and Tamil.
- Augmented Reality (AR) Produce Sizing:** Prototype AR camera overlay allowing buyers to inspect average fruit and vegetable dimensions before placing bulk orders.

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5. Promotional Deliverables

5.1 Professional Branding & Identity

- Brand Name:** Farmora (Combining *Farming* + *Flora* + *Agora / Marketplace*)
- Tagline:** *"Connecting Soil to Shelf ? Fair, Fresh, and Fast."*
- Primary Color:** Agri Green (‘#2E7D32‘) paired with Golden Harvest (‘#F57C00‘).

5.2 A3 Promotional Poster Blueprint

- Format:** High-resolution A3 Poster (Attached in ‘docs/academic_submission/milestone_05/FARMORA_A3_POSTER.md‘).
- Visual Composition:**
- Top Hero: Striking split graphic of a happy rural farmer holding fresh produce next to a delivery truck and supermarket shelf.
- Value Callouts: 0% Middleman Cut | Real-time Cold-Chain Tracking | 48-Hour Guaranteed Escrow.
- QR Call to Action: Large scannable QR linking to the live app (‘https://farmingapp-24b34.web.app‘) and APK download.

5.3 AI-Generated Promotional Videos (Scripts & Production Plans)

Two targeted 90-second video production blueprints (with narration scripts and scene breakdowns) are detailed in: ‘docs/academic_submission/milestone_05/VIDEO_SCRIPTS_AND_PROMOTION.md‘:
1. **Video 1 (Technical Demonstration - 90s):** Targeted at technical evaluators and software engineers. Demonstrates the Flutter architecture, Firebase Realtime streams, Cloudinary video integration, and QR batch scanning.
2. **Video 2 (Business Pitch & Investor Pitch - 90s):** Targeted at venture investors and agricultural stakeholders. Demonstrates the Sri Lankan post-harvest crisis, addressable market size, unit economics, and societal impact.

Farmora Investor & Academic Pitch Deck

Connecting Soil to Shelf ? Fair, Fresh, and Fast

SLIIT Department of Software Engineering | Milestone 5 & Final Evaluation

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Slide 1: Title Slide

- Title:** Farmora
- Subtitle:** The Next-Generation Smart Agricultural Marketplace & Traceable Cold-Chain Ecosystem
- Presented by:** Team Farmora (Ahsan Mohammed, Kajana S., Sureka)
- Institution:** Sri Lanka Institute of Information Technology (SLIIT)

Slide 2: The Agricultural Crisis in Sri Lanka

- The Problem:**
- 40% Loss to Middlemen:** Fragmented broker cartels at physical economic centres exploit smallholder farmers.
- 30% Post-Harvest Spoilage:** Lack of verified cold-chain logistics causes fresh perishables to rot during transit.
- Zero Traceability:** Supermarkets and hospitality buyers cannot verify chemical safety, harvest date, or origin.
- Payment Defaults:** High risk of bad debts and delayed cash disbursements.

Slide 3: The Solution ? Farmora Ecosystem

- Direct Digital Marketplace:** Farmers sell directly to supermarkets, hotel syndicates, and bulk buyers.
- Field Video Verification:** Farmers upload 10-second proof-of-freshness video directly from the field.
- Cryptographic Batch QR:** Unique QR codes attached to crates verify harvest date, farmer identity, and agronomy data.
- Guaranteed Escrow Protection:** Funds locked at order placement; released within 2 seconds of verified delivery.
- Integrated Cold-Chain Fleet Hub:** Real-time IoT temperature monitoring across all transport reefers.

Slide 4: Multi-Role Experience

- For Farmers:** 1-tap listing, voice search, market price ticker, instant escrow payouts.
- For Wholesale Buyers:** Transparent grades, field video inspection, bulk procurement, dispute resolution.
- For Fleet Transporters:** Route matching, vehicle & driver depot management, live GPS, digital POD.
- For Admin / Regulators:** Escrow treasury settlement, user KYC validation, dispute resolution tribunal.

Slide 5: Technical Excellence & Architecture

- Front-End:** Flutter 3.x (Multi-platform compilation to Android Release APK & Progressive Web App).
- Back-End:** Firebase Serverless (Cloud Firestore real-time streams, Cloud Storage, Authentication).
- Media Engine:** Cloudinary API for optimized video streaming and automatic cleanup upon order fulfillment.
- Security:** Granular Role-Based Access Control (RBAC) enforced at Firestore engine level.
- Offline First:** Local caching enables uninterrupted harvest logging in low-connectivity rural farms.

Slide 6: Usability & Testing Validation (Milestone 3)

- Usability Benchmark:** System Usability Scale (SUS) Score of **84.5 / 100 (Grade A)**.
- Task Success Rate:** Over 95% across critical user paths.
- Accessibility:** High-contrast daylight UI (WCAG 2.1 AA) with full trilingual support (Sinhala, Tamil, English).

Slide 7: Business Model & Revenue Streams

1. **Marketplace Commission:** 2.5% take rate on wholesale transactions.
2. **Escrow Guarantee Fee:** 1.0% platform assurance fee.
3. **Logistics Dispatch Matching:** 3.5% freight matching commission.

4. **Enterprise SaaS:** $25/month for supermarket inventory API synchronization.
   - Total Addressable Market (TAM):** $420M+ agricultural trade volume in Sri Lanka.

Slide 8: Go-to-Market & Roadmap

- Q1:** Pilot rollout in Dambulla & Nuwara Eliya agricultural belts (250 verified farmers).
- Q2:** Formal procurement integration with supermarket chains (Keells, Cargills).
- Q3:** Full logistics fleet matching rollout across North & Central Provinces.
- Q4:** Regional expansion and agri-microcredit underwriting partnerships.

Slide 9: The Deliverables

- Live Progressive Web App:** https://farmingapp-24b34.web.app
- Production Android APK:** ‘build/app/outputs/flutter-apk/app-release.apk‘ (99.3MB, compileSdk 36)
- Open GitHub Repository:** https://github.com/AHSANMOHAMMED/Farmora-App.git
- Complete Project Report:** Milestones 1 through 6 documentation.

Slide 10: Conclusion & Q&A

- "Farmora connects soil to shelf ? ensuring fair profits for farmers, guaranteed freshness for buyers, and zero wasted food."*
- Thank you! We welcome your questions.**

Farmora Promotional Assets Blueprint (Milestone 5)

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1. Professional Brand Identity

- Product Name:** Farmora
- Official Tagline:** *"Connecting Soil to Shelf ? Fair, Fresh, and Fast."*
- Typography:** Google Poppins (Headings) + Inter (Body) + Noto Sans Sinhala & Tamil.
- Color Palette:**
- Agri Green: ‘#2E7D32‘
- Fresh Sprout: ‘#4CAF50‘
- Amber Harvest: ‘#F57C00‘
- Clean Cloud: ‘#F8FAFC‘

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2. A3 Promotional Poster Layout Spec (Print Ready)

‘‘‘
??
? FARMORA ?
? Connecting Soil to Shelf ? Fair, Fresh, and Fast ?
??
? ?
? [HERO VISUAL: FARMER ? REEFER TRUCK ? BUYER] ?
? "Empowering 1.5 Million Farmers with Direct Digital Commerce" ?
? ?
??
? ?
? [ICON: 0% BROKERS] [ICON: IOT COLD CHAIN] [ICON: SECURE ESCROW]?
? Cut Out Middlemen Real-Time Telematics 100% Guaranteed Pay ?
? Earn up to 40% higher Monitor temperature Automated release ?
? farm-gate revenues to eliminate spoilage upon QR inspection ?
? ?
??
? ?
? SCAN TO EXPERIENCE THE FUTURE OF AGRI ?
? ?
? ????????????????????????????? ?
? ? [QR CODE] ? ?
? ? https://farmingapp-24b34. ? ?
? ? web.app ? ?
? ????????????????????????????? ?
? ?
? Available on Android (APK) & Progressive Web App ?
? ?
??
? SLIIT Faculty of Computing | Department of Software Engineering 2026 ?
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‘‘‘

3. Video 1: Technical Demonstration (Duration: 90 Seconds)

Intended Audience: Technical Evaluators, Software Architects, Examination Panel
Tone: Informative, Professional, Architecture-Focused

- [0:00 - 0:15] Architecture Overview:**
- Visual:* Clean high-level architecture diagram illustrating the Flutter client connecting to Firebase Cloud Functions, Firestore real-time listeners, and Cloudinary media processing.
- Voiceover:* "Welcome to Farmora. Built on Flutter and Firebase serverless architecture, Farmora creates an ultra-responsive, real-time bridge connecting producers, wholesale buyers, and logistics networks across Sri Lanka."
- [0:15 - 0:40] Farmer Flow & Barcode Generation:**
- Visual:* Screen recording of the Farmer adding a harvest. Showing the live camera capturing fresh produce video, uploading to Cloudinary, and instantly generating an encrypted batch QR code.
- Voiceover:* "Farmers record video proof of freshness right in the field. Our

system generates a cryptographic QR batch identity that tracks crop grade, harvest timestamp, and soil parameters."
- [0:40 - 1:10] Buyer Escrow & Cold-Chain Telematics:**
- Visual:* Buyer browsing the catalog, streaming the harvest video preview, and depositing funds into escrow. Screen splits to the Transporter Fleet Hub with live Google Maps tracking and real-time cold-storage temperature graphs.
- Voiceover:* "Buyers inspect harvest video directly before purchase. Escrow locks payment until delivery. In the Fleet Hub, IoT telemetry streams reefer temperatures in real-time, preventing transit spoilage."
- [1:10 - 1:30] Proof-of-Delivery & Escrow Release:**
- Visual:* Transporter captures recipient signature and delivery photo. The order status updates instantly via Firestore snapshot, and escrow funds are immediately transferred to the farmer’s balance.
- Voiceover:* "Upon QR scan and photographic Proof-of-Delivery, escrow funds release automatically in under two seconds. Farmora is built for performance, security, and true multi-platform scalability."

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4. Video 2: Business & Investor Pitch (Duration: 90 Seconds)

Intended Audience: Venture Investors, Agricultural Policy Makers, Business Evaluators
Tone: Inspiring, Urgent, High Energy, Commercial Focus

- [0:00 - 0:20] The Problem:**
- Visual:* Documentary footage of Sri Lankan vegetable markets, overflowing crates, and disappointed farmers negotiating with brokers.
- Voiceover:* "In Sri Lanka, over 30% of harvested vegetables rot before reaching the shelf, while hardworking farmers lose up to 40% of their income to predatory intermediaries. Food security is at risk."
- [0:20 - 0:45] The Solution:**
- Visual:* Fast-paced cinematic showcase of Farmora on mobile devices: vibrant green UI, 1-tap ordering, transparent price comparison.
- Voiceover:* "Meet Farmora ? the digital operating system for agricultural commerce. We bypass broker cartels, directly connecting smallholders with supermarkets, hotels, and bulk buyers through video-verified trading."
- [0:45 - 1:15] Business Model & Traction:**
- Visual:* Infographics displaying the 2.5% marketplace commission, 3.5% logistics dispatch fee, and expanding TAM across South Asia.
- Voiceover:* "Our multi-sided business model captures a 2.5% transaction commission, freight matching fees, and enterprise supply chain SaaS. With over 1.5 million farmers across Sri Lanka, Farmora targets a $420 million digital trade pipeline."
- [1:15 - 1:30] Call to Action:**
- Visual:* Farmer smiling in a lush green field holding a smartphone, with the Farmora logo and download link.
- Voiceover:* "Fair pricing for farmers. Guaranteed quality for buyers. Zero wasted food. Join us in transforming agriculture. Farmora ? connecting soil to shelf."

SLIIT Department of Software Engineering - Faculty of Computing

Mobile Application Project: Farmora

Milestone 6: Individual Contribution Logs & VIVA Preparation (All LOs)

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1. Team Overview & Git Verification

- Project Repository:** ‘https://github.com/AHSANMOHAMMED/Farmora-App.git‘
- Primary Branches:** ‘main‘, ‘dev/swami‘, ‘Kajana‘, ‘suka‘, ‘feature/ahsan-dev‘

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2. Individual Contribution Logs

2.1 Member 1: Ahsan Mohammed (Team Lead & Full-Stack Architect)

- Designated Responsibilities:**
- System architecture, Flutter Clean Architecture setup, Firebase Security Rules design.
- Escrow payment lifecycle, PayHere integration, and multi-role administrative dispute moderation.
- Harvest Barcode / QR cryptographic generation, Cloudinary video integration, and automated storage lifecycle.
- Android release pipeline (Java 21 LTS, Gradle build scripts, ProGuard optimization) and Firebase Web PWA deployment.
- Milestone-by-Milestone Contributions:**
- M1:* Formulated architecture specifications (‘FARMORA_FULL_BUILD_SPEC.md‘), defined persona requirements for Wholesale Buyers and Admins.
- M2:* Built core theming engine (Material 3), multi-role navigation controllers, and responsive layout foundations.
- M3:* Conducted secure design audit on Cloud Firestore rules; remediated RBAC vulnerabilities for financial settlements.
- M4:* Implemented buyer catalog, escrow locking/unlocking engine, video compression/streaming client, and release APK compilation.
- M5:* Designed business pitching deck, financial revenue models, and video pitch storyboards.
- M6:* Prepared comprehensive architecture documentation and VIVA defense guides.
- Evidence / Git Commits:**
- Commit ‘4b1395b‘: ‘fix(fleet): add check for empty branches before adding vehicle or driver‘
- Commit ‘55bafbc‘: ‘fix: address Copilot review ? security, rules, timestamps, tracking‘
- Commit ‘edfa795‘: ‘feat(admin): add Security Audit Trail, Treasury Escrow Settlements‘

2.2 Member 2: Kajana S. (Farmer Operations & Crop Management UX)

- Designated Responsibilities:**
- Farmer dashboard, harvest creation forms, and crop health modules.
- AI plant disease scanner client and agronomic tips integration.
- Profile customization, localized language strings (Tamil/English), and farm plot management.
- Milestone-by-Milestone Contributions:**
- M1:* Executed contextual inquiries with vegetable farmers; mapped the step-by-step harvest listing flow.
- M2:* Designed high-fidelity prototypes for the Farmer Home dashboard and field crop ledger.
- M3:* Moderated usability testing sessions with rural participants; gathered feedback on camera/video usability.
- M4:* Implemented ‘add_product_screen.dart‘, plot ledger UI, and agricultural advice feed.
- M5:* Co-authored the Go-to-Market strategy focusing on agrarian community hubs in Jaffna and Central Province.
- M6:* Prepared contribution log and VIVA slides on agricultural UX patterns.
- Evidence / Git Commits:**
- Commit ‘715f288‘: ‘Update profile screen‘
- Branch: ‘Kajana‘ / ‘kajana_task‘

2.3 Member 3: Sureka (Suka) (Logistics & Transporter Fleet UX)

- Designated Responsibilities:**
- Transporter portal, dispatch tracking, and cold-chain monitoring.
- Fleet management bottom sheets (Branch Depots, Vehicles, and Driver registration).
- Telematics mapping views (Google Maps Flutter integration) and Proof-of-Delivery (POD) captures.
- Milestone-by-Milestone Contributions:**
- M1:* Conducted research on third-party agricultural freight operators; identified deadhead mileage issues.
- M2:* Prototyped interactive fleet telematics dashboards and cold-storage alert cards.
- M3:* Tested fleet registration flows; identified edge cases where empty branch lists caused unhandled UI exceptions.
- M4:* Built ‘logistics_fleet_hub_screen.dart‘, ‘vehicle_maintenance_sheet.dart‘, and GPS location listeners.
- M5:* Designed marketing materials targeting freight fleet owners and transport logistics syndicates.
- M6:* Prepared VIVA defense documentation regarding Google Maps telematics and cold-chain compliance.
- Evidence / Git Commits:**
- Commit ‘9dc6103‘: ‘transportor ui‘
- Branch: ‘suka‘

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3. Individual Self-Reflection Templates (One-Page Signed Form)

(Each student will print, review, sign, and hand over to the evaluation panel during VIVA).

Template Structure:

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INDIVIDUAL SELF-REFLECTION REPORT

Department of Software Engineering, SLIIT

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Student Name: _____ Student ID: _____

Assigned Role: _____ Group Name: Farmora

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1. Specific Contributions Across Milestones:
[Detailed breakdown of personal architectural, coding, UI, or testing tasks]

2. Technical & Practical Challenges Faced:
[How issues such as empty state errors, multi-platform builds, or Firebase rules were solved]

3. New Skills Learned & Applied:
[Flutter/Dart state management, asynchronous streams, IoT cold-chain concepts, Material 3]

4. What I Would Do Differently in Retrospect:
[e.g., Implementing end-to-end integration tests earlier, conducting wider field trials]

Student Signature: _____ Date: _____

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4. Comprehensive VIVA Defense & Q&A Preparation

Be prepared to answer these critical questions during the individual and group evaluation:

Q1: Why did you choose Flutter over React Native or Native development?

> **Answer:** Flutter compiles directly to native ARM machine code using the Impeller/Skia engine, eliminating the JavaScript bridge bottleneck. This ensures consistent 60+ FPS animations during heavy Google Maps telematics and camera barcode scanning. Crucially, Flutter allowed our team to maintain a single unified codebase that runs seamlessly as an Android APK and a Progressive Web App (PWA) with zero code duplication.

Q2: How does the system protect farmer funds and prevent buyer fraud?

> **Answer:** We implemented a two-sided Escrow payment engine. When a buyer orders, funds are deposited into a secure escrow account managed via Cloud Firestore security rules. Payment is never released immediately. Only after the transporter delivers the goods, the buyer scans the tamper-evident batch QR code, and photographic Proof-of-Delivery is submitted, the escrow unlocks and credits the farmer within seconds.

Q3: How do you handle cold-chain perishable failures during transit?

> **Answer:** The Transporter Fleet Hub streams IoT temperature sensor data directly into Firestore. If refrigerated truck temperature breaches safe agricultural thresholds (e.g. rising above 7℃ for leafy greens), the system triggers an immediate push notification alert and logs a permanent breach flag on the batch ledger, allowing the buyer to inspect or dispute before escrow settlement.

Q4: How is data synchronized in rural areas with poor cellular coverage?

> **Answer:** We configured Cloud Firestore with offline persistence enabled. When farmers or drivers enter rural valleys without internet, they can continue recording harvests, logging fleet records, and taking POD photos locally. The Flutter client queues mutations in local cache and automatically syncs to the cloud the moment network connectivity resumes.